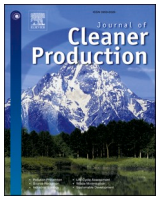
Journal of Cleaner Production 437 (2024) 140632



Contents lists available at ScienceDirect

Journal of Cleaner Production

journal homepage: www.elsevier.com/locate/jclepro





Understanding the food waste behaviour in university students: An application of the theory of planned behaviour

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ARTICLE INFO

Handling Editor: Giovanni Baiocchi

Keywords:
Food waste
Food waste behaviour
Extended theory of planned behaviour
Regression
Correlation
Attitude
Morals
Knowledge

ABSTRACT

Food waste has a significant impact on the environment, national economies, and food security, necessitating appropriate action by policymakers. Food waste is prevalent among the young generation, so it becomes prudent to understand their food waste behaviour, involving the identification and evaluation of significant variables. This study seeks to comprehend their food waste behaviour as well as their intent to engage in such behaviour. An extended theory of planned behaviour was utilized to achieve the objectives, and a conceptual model and hypotheses were formulated accordingly. Some extensive constructs were incorporated with the theory of planned behaviour's major constructs. Data was collected via a questionnaire survey from a sample of 345 students at the University of Kashmir, Srinagar, India to identify relevant factors. Correlation analysis and regression analysis (linear regression model) were used to better understand the relationship between various variables. The intention of students not to waste food was successfully predicted using the extended theory of planned behaviour. Our study's findings revealed that several factors, such as one's attitude, behavioural control, knowledge about food waste, moral standards, eating habits, and leftover reuse routine, have a significant influence ($p < 0.01$) on students' desire to avoid food waste and explain 68.6% of the variance in intention to not waste food. Knowledge about food waste was by far the most significant factor associated (with max. β coefficient) with the desire not to waste food. Subjective norms did not reach statistical significance ($p > 0.01$) and could therefore not be used to predict intentions. The research highlights effective strategies to reduce food waste among youth by outlining the most impactful factors that explain the intention to waste less food and thus contribute to addressing the global issue of food waste.

1. Introduction

1.1. Background and objectives

Food waste has grown to be a major issue throughout history in both developed and developing countries (Bhatti et al., 2023; Khan et al., 2022). An estimated 1.3 billion tonnes of food are wasted or lost worldwide each year. This corresponds to one-third of the food created for human consumption (Gustavsson et al., 2011). Due to the resource-intensive nature of food production, food losses, and wastes have a wide range of indirect negative effects on the environment, including soil erosion, deforestation, water and air pollution, as well as greenhouse gas emissions from the processes of food production, storage, transportation, and waste management (Mourad, 2015). We consume food in an unsustainable way, endangering both our community and our resources. Food waste generation at the student level is a complicated and diverse problem driven by several factors and behaviours (Wang et al., 2023). Various food waste drivers are examined to understand the effective policy strategies for managing food waste sustainably. Food waste prevention has gained worldwide attention, with Sustainable Development Goal 12.3 seeking to reduce global food waste per person by half by 2030. Reaching this goal would be a big contribution to the fight against climate change (FAO, 2022).

In India, food is wasted as much as is available for consumption to the whole of United Kingdom as reported by the Corporate Social Responsibility (CSR) journal and as a result, millions of people are sleeping hungry in a population of 1.4 billion (Chowdhuri, 2020). In the Global Hunger Index-2023, India ranks 111th out of the 125 countries. Food

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https://doi.org/10.1016/j.jclepro.2024.140632
Received 1 August 2023; Received in revised form 18 December 2023; Accepted 5 January 2024
Available online 5 January 2024
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waste per person per year in India in the high-income group, middle-income group, and poor sectors, respectively stands at 90 kg, 68 kg, and 63 kg, according to the United Nations Environment Programme's Food Waste Index Report 2021(Chaudhary et al., 2021; Sahoo et al., 2023). The amount of food waste per person in Europe and North America is between 95 and 115 kg per year, whereas in sub-Saharan Africa and South/Southeast Asia, it is approximately 6–11 kg per year (Gustavsson et al., 2011). A study by Thi et al. (2014) found that developed countries waste an average of 107 kg of food per person per year while developing countries waste an average of 56 kg per person per year. These figures indicate that there is a significant difference in food waste generation between developed and developing countries, with higher living standards leading to greater food waste (Brian et al., 2013).

In the Kashmir valley, food waste is a serious problem, especially during major events like weddings, parties, and get-togethers. It has become customary to offer a range of treats at weddings. Foods of different varieties and cuisines are prepared and served to guests. More food is prepared than one can eat, hence the excess is finally thrown away even though people in Kashmir carry the leftovers with them. To repair the damage, we must learn how to produce and consume food in a sustainable way. Therefore, it's crucial to comprehend consumer behaviour in relation to food consumption if we intend to promote less wasteful practices to achieve sustainability (Lazell, 2016). It is crucial to have in-depth knowledge of the variables influencing consumer intention and behaviour concerning food waste at the time of purchase and during the preparation at home (Salins and Aithal, 2022).

Perusal of the literature shows that research on food waste behaviour among the young generation is lacking in the Kashmir Valley. Our central motivation for piloting this study involved minimizing the existing gap between the relevance of food waste research and the main behavioural driving forces that generate food waste. We decided to identify and evaluate the significant variables that influence and determine the student's food waste behaviour to mitigate the negative effects of consumption. This was done with the application of the extended theory of planned behaviour (ETPB) to examine the food waste practices of Kashmir University students as well as to identify and comprehend the underlying factors that support or restrict such specific food waste practices and to investigate how students' attitudes, morals, and knowledge have a role in that behaviour. This study also aims to determine the interrelations between each of the theory's "classical components," namely Intention, Attitude, Social Norms, and Perceived Behavioral Control, and as well as the extended variables (knowledge about food waste, moral norms, purchasing behaviour, eating habits, and leftover reuse routines). The interventions will influence the students' behaviours by helping them to reduce their food waste, and in turn, will help develop prevention strategies for food waste and will play a major beneficial impact on the economy, society, and the environment (Fan et al., 2022). Our objective is to analyse how universities can reduce FW levels, and our main contribution is to suggest where actions should be taken. Gaining a comprehensive understanding of the current state of research in this particular area can provide valuable insights for policymakers, practitioners, and researchers, allowing them to make informed decisions regarding optimal strategies, potential obstacles, and avenues for future exploration".

1.2. Review of literature

1.2.1. Food waste and universities

According to FAO (2022), Food waste (FW) is defined as any food for human consumption that is discarded after the expiry date or left to spoil. Food waste is defined as edible food that is meant for human use but is ultimately not utilized for any reason (Van Alboom, 2021). FW at universities is caused by dining hall preparation and consumer behaviour. Deliberador et al. (2021a,b) identified various reasons such as food quality, portion size, and attitudes. Seasonal influence on FW is overlooked. Qian et al. (2022a) found FW is influenced by factors including gender, education, and awareness. The spatial characteristics of a region have a significant impact on the amount of food waste (FW) generated. A recent survey conducted by Qian et al. (2022b) at Chinese universities found that regional economic development plays a crucial role in FW generation in China. According to the research, regions with a lower carbon food footprint, such as Northeast and Northwest China, have less demand for rice and meat, and instead, wheat is the staple food. This can be attributed to the fact that people in the southern regions are more prosperous and have higher purchasing power, leading to higher levels of consumption and waste of rice and meat. This finding is consistent with a previous study conducted by Qian et al. (2022a). In another study, Zhang et al. (2021) pointed out that FW per meal, per capita, is influenced by a range of factors, including personal attributes such as age and gender, and other factors related to consumption behavior. These factors include excessive purchasing, tasteless food, personal preference for gourmet over staple food, undesired accompaniments, selection of disposable tableware, cheap food, and large-sized portions. According to a recent international study conducted by Leal Filho et al. (2021), out of 52 higher education institutions, 60% of the universities do not keep track of the amount of food waste generated by their canteens. This lack of measurement makes it difficult to fully comprehend the extent of food waste production, its impact on waste and related costs, and the need to dispose of it. By quantifying the amount of food waste produced, universities can identify and implement preventive measures, identify operational inefficiencies, and set rational goals for improvement. Therefore, it is essential that universities start measuring their food waste to promote efficiency and sustainability. Martinho et al. (2022) conducted a study on food waste generated in a Portuguese university. They found that plate waste, which refers to the food left uneaten on the plates, accounted for 86.7% of all food waste generated. This indicates that the behavior of the students plays a significant role in the amount of food waste produced in the university.

Grech et al. (2020) proposed reducing package use and recycling to minimize food waste in the universities. The trayless initiative implements this idea as an infrastructural and cultural action (Davison et al., 2022). The implementation of sustainable practices in specific areas such as ethical procurement and sustainable food consumption strategies has been described by some authors as a crucial step towards changing students' behavior, while considering cultural differences. Educational strategies have been developed to promote sustainable practices among students and the university community, specifically focusing on food waste reduction. It is a fact that governments and universities are committed to embedding sustainability into institutional activities such as teaching and research. They also encourage students to study sustainability topics, collaborate with the community, provide staff development, and reduce carbon emissions. Campuses are being transformed to incorporate sustainability practices in all aspects, including resources such as water, grounds, food waste, procurement, and purchasing. This is a long-term commitment to sustainability that universities and governments take seriously (Cavazos et al., 2023; Davison et al., 2022; Grech et al., 2020; Martinho et al., 2022).

Numerous factors like cultural, political, economic, geographic, and sociodemographic have been studied with reference to their role in FW behaviour (Thyberg and Tonjes, 2016). Therefore, we attempted to address this problem from the perspective of the theory of planned behaviour to get a comprehensive and holistic answer.

1.2.2. Food waste and TPB

Understanding the variables that influence food waste behaviour and understanding consumer behavior about food before it becomes waste, during consumption, or when it is "in use" is necessary to reduce food waste (Lazell, 2016). To look into the hidden nature of the food waste problem, consumer behaviours, routines, and habits can be used to better understand consumer food waste behavior (Lazell, 2016). The Theory of Planned Behavior (TPB) makes predictions about a person's

intention to act in a certain way at a particular time and place (Werf et al., 2019; La Barbera et al., 2022). People decide how they will behave in advance, act on that decision, and think about the conclusion they want to attain, as a result, behaviours are influenced by certain intentions (Aydin and Aydin, 2022). It is a Framework that allows for extensions, ultimately improving behaviour prediction. It contends that behaviour is caused by a person's goals, which are a function of three primary variables, including their attitude towards behaviour, their perception of behavioural control, and subjective norms (Ajzen, 1991), and some extensive variables like purchasing behavioural control, knowledge about food waste, morals norms, eating habits, intention not to food waste, leftover reuse routine (Graham-Rowe et al., 2015). Compared to the original TPB, the expanded TPB is better at predicting food waste behaviour (Van Alboom, 2021).

1.2.3. Comprehension of variables

1.2.3.1. Attitude about food waste. Degree of assessment or evaluation of a person by favourable or even unfavourable in terms of the behaviour (Ajzen, 1991). Based on one's attitudes about the behaviour, one is held to believe that the individual is highly motivated and capable of thinking deliberately (a deliberative process)-a process whereby attitudes affect behaviour. In the TPB, intentions are taken to be a clear indication of how much effort or how hard people are willing to try to engage in the behaviour (Conner and Armitage, 1998). An individual's intention to not waste food is positively influenced by positive attitudes toward food waste (Blešić et al., 2021).

1.2.3.2. Subjective norms. These refer to the perceived social pressure to act in a certain way or refuse to act in a certain way (Ajzen, 1991). Subjective norms by way of societal pressure influence an individual to engage in particular conduct. Some research that examined food waste in families demonstrated that subjective norms have no influence on the intention to waste food (Chen, 2023). If one's close friends and family support reducing food waste, their intention to do so will likely be stronger. The subjective norm strengthens the purpose of managing food waste, which has the biggest impact on modifying behaviour to reduce food waste. This demonstrates that outward social pressure impacts people's intentions more than internal pressure (Ariyani and Ririh, 2020; Razali et al., 2020; Jereme et al., 2016).

1.2.3.3. Perceived behavioural control. Those who felt they had more control over their food waste behaviour were less likely to throw out more food than those who felt they had less control (Ajzen, 1991). The theory of Planned Behaviour claims that when people feel they have enough control, their intentions to engage in a certain behaviour increase (Stancu et al., 2016; Aydin and Aydin, 2022).

1.2.3.4. Purchasing behaviour. Purchasing behaviour has two significant factors that influence how much food is wasted. Individuals who give careful consideration before purchasing food and buy the exact amount of food they need also help to reduce food waste (Aydin and Yildirim, 2020). When people only shop at big supermarkets, the quantity of food that is wasted is largest. When people buy at other stores, small shops, and local markets, the amount of food that is wasted is the lowest (Jörissen et al., 2015).

1.2.3.5. Knowledge about food waste. With the highest level of reported knowledge of the effects of food waste, purchase plans are essential, and consumers are increasingly planning more of their meals and food purchases (Richter, 2017). Reduced amounts of student's food waste were associated with awareness of the consequences of food waste and the ability to interpret the information on food labels (Hatab et al., 2022). Students who knew how to freeze or preserve food tend to throw food away less frequently (Heng and House, 2022).

1.2.3.6. Moral norms. The feeling of guilt when discarding food is defined as a moral norm (Wong, 2019; Lavén, 2017; Parizeau et al., 2015; Quested et al., 2013). Moral norms are a good indicator of intention, which is affected by a sense of moral obligation and independently predicted attitudes. When people believe that wasting food is against his/her morals, they avoid doing so (Aydin and Yildirim, 2020). The feeling of guilt (moral norms) is determined as the most important variable for the intention to not waste food (Soorani and Ahmadvand, 2019).

1.2.3.7. Intention to not waste food. Behaviour is driven by intention, which is defined as the willingness and desire to carry out a particular behaviour (Wong, 2019; Ajzen, 2011). As predicted, a strong determinant of food waste behaviour is the intention to reduce it. The intention to reduce food waste is considered a significant decision-making factor (Anjaka, 2022). Avoiding food waste and Food waste generating behaviour are mostly determined by intention (Kritikou et al., 2021).

1.2.3.8. Eating habits. Eating at home facilitates the adoption of food waste prevention practices (Savelli et al., 2020). The frequency of food waste increases when people eat outside the home and use expiration dates and appearance to decide if food is spoiled (Barone et al., 2022). Due to spoilage, the limited shelf life of fresh food, and individuals forgetting about food left in the fridge, a significant amount of food is wasted (McCarthy and Liu, 2017).

1.2.3.9. Leftover reuse routine. The aspects of food waste that are most clearly manifested are customer non-re/use of food leftovers (dos Santos et al., 2022). When it comes to measures to avoid food waste, eating leftovers seems to be an important factor in reducing food waste (Bravia et al., 2020).

1.3. Research hypotheses

The extended theory of planned behaviour (ETPB) has been used in this study to predict university students' behaviour (Fig. 1). The hypotheses were formulated and proposed as a model of ETPB. It is acceptable to use behavioural models to describe how young people behave when it comes to food waste (Quested et al., 2013).

1.3.1. Attitude towards food waste

Based on one's attitudes about the behaviour, one is held to believe that the individual is highly motivated and capable of thinking deliberately (a deliberative process)-a process whereby attitudes affect behavioural intention (Koivupuro et al., 2012; Wang et al., 2021). Thus,

Hypothesis 1. Attitude has a positive effect on the intention not to waste food.

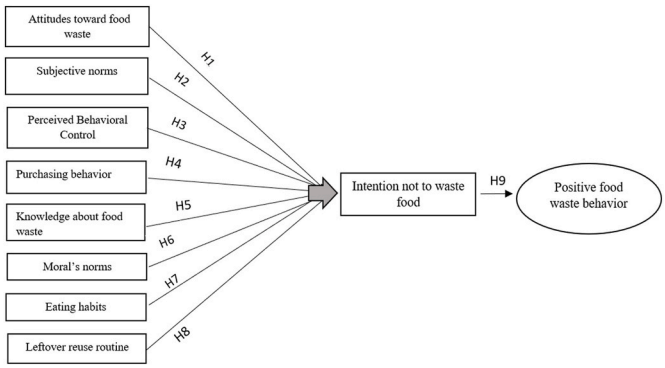


Fig. 1. ETPB conceptual model.

1.3.2. Subjective norms

If one's close friends and family support reducing food waste, their intention to do so will likely be stronger (Ajzen, 1991). Hence,

Hypothesis 2. Subjective norms have a positive effect on the intention not to waste food.

1.3.3. Perceived behavioural control

Those who felt they had more control over their food waste behavior were less likely to throw out more food than those who felt they had less control (Ajzen, 1991; Palmieri and Palmieri, 2022).

Hypothesis 3. Perceived behavioural control has a positive relationship with an intention not to waste food.

1.3.4. Purchasing behavior

To reduce food waste, pre-shopping planning using predefined shopping lists is an effective strategy. The more robust the pre-shopping planning, the more will be the intention not to waste food (Zainal and Hassan, 2019).

Hypothesis 4. Purchasing behaviour has a positive relationship with an intention not to waste food.

1.3.5. Knowledge about food waste

The amount of FW by households would likely reduce if people were aware of the consequences of food waste (Aka and Buyukdag, 2021) and had the necessary skills to interpret and understand the information on food labels.

Hypothesis 5. Knowledge has a positive relationship with an intention not to waste food.

1.3.6. Moral norms

The less food each person feels obligated to throw away, the more likely it is that household waste will decrease (Ajzen, 1991).

Hypothesis 6. Moral norm has a positive relationship with an intention not to waste food.

1.3.7. Eating habits

Eating at home makes it easier to adopt measures to reduce food waste (Savelli et al., 2020). The amount of food that is wasted may be significantly influenced by individuals' ability to finish the meal that is served to them, more they finish the plate more will be increased in their intention to not food waste (Aydin and Yildirim, 2020).

Hypothesis 7. Eating habits have a positive effect on the intention not to waste food.

1.3.8. Leftover reuse routine

Reusing the leftover can reduce food waste, individual having strong leftover reuse routines then they have a high level of intention not to waste food (Zainal and Hassan, 2019).

Hypothesis 8. Leftover reuse routine has a positive effect on the intention not to waste food.

1.3.9. Intention not to waste food

People who have a stronger intention to reduce food waste tend to throw away less food (Zainal and Hassan, 2019).

Hypothesis 9. Intention not to waste has a positive effect on Positive food waste behaviour.

2. Material and methods

2.1. Data collection and sample size

The study aimed to gain insights into student's attitudes and perceptions regarding food waste and to classify them based on their attitudes. To achieve this, a 35-item questionnaire was created using ETPB as a conceptual framework (Ajzen, 1991), and it contained questions from widely used and previously validated household/consumer food waste surveys. For the collection of data from all 48 departments of the University of Kashmir, stratified proportionate random sampling was used. The sample size was determined by Sloven's formula (Eq. 1), and 345 students undergoing post-graduate programs participated in the study. Along with collecting socio-demographic information, the survey included questions about food purchasing habits, as well as attitudes and everyday behaviours related to food waste. Respondents provided their answers using a five-point Likert scale, ranging from "do not agree at all" (1) to "absolutely agree" (5). The validity of the data was tested using factor analysis, and for confirmatory tests, correlation, and regression analysis were undertaken.

Sloven's formula is calculated as (Eq. (1)):

$$n = \frac{N}{(1 + Ne^2)} \quad \text{Eq.1}$$

where:
n: Sample size needed.
N: Population size
e: Acceptable margin of error.

2.2. Reliability of the questionnaire

The Statistical Package for Social Sciences (SPSS) 26 version was used to code, compute, and analyse the data after eliminating the redundant surveys. Using Cronbach's alpha, the questionnaire's internal consistency was evaluated. According to Nunnally (1978), Cronbach's α values greater than 0.70 suggest greater reliability and higher internal consistency in the measured dimension. The total consistency of the entire questionnaire (35 items), according to the results of reliability using the Cronbach alpha approach for internal consistency was 0.84 showing high data reliability (Table 1). Additionally, the study's Cronbach's alpha for each component is higher than 0.70. It is evident that the survey used in this research has high internal consistency, and as a result, statistical analysis and interpretation of the results were applied to it with confidence.

2.3. Statistical analysis

After the reliability, exploratory factor analysis was done to reduce the dimensionality of the data. Factor analysis is often used to explain a larger set of measured variables with a smaller set of latent constructs. KMO & Bartlett's test of sphericity, Principal Component Matrix, scree plot used, rotated component matrix were all the steps needed for exploratory factor analysis. The factorability of the data is evaluated using Bartlett's test of Sphericity and the Kaiser-Meyer-Olkin measure of sample adequacy. To investigate the multicollinearity of the variables, the determinant score was generated. Kaiser's Criterion and the Scree

Table 1
Cronbach's alpha coefficient– (reliability).

Variables	Cronbach's α
Purchase_Behavior	0.78
Attitude	0.80
Subjective_Norms	0.88
Perceived_Behavioural_Control	0.70
Knowledge_about_Food_Waste	0.80
Eating_Habits	0.73
Moral_Norms	0.75
Leftover_Reuse_Routine	0.88
Intention_not_to_Waste_food	0.80
Positive _Food_Waste_Behavior	0.80
Overall (35) items	0.84

test were looked at to determine how many variables need to be removed. To reduce the number of variables that have large loadings on each factor, the Varimax orthogonal factor rotation approach was used. After reducing the dimensionality of the data, descriptive statistics of factors were done followed by correlation and regression analysis to get the outcome.

3. Results

3.1. Descriptive characteristics of respondents

To investigate the food waste behaviours of the students, a preliminary descriptive analysis was conducted. Fig. 2 illustrates the frequency of food waste reported by participants across various food categories. Rice (46%) contributes more to the food waste stream than other food wastes. Meat and fish were found to be wasted the least. Approximately 79% of the participants believed they hardly wasted any meat and fish. In contrast, the fruit and vegetable category, as well as the bread and bakery category, were identified as the most wasted food categories after rice.

The respondents were fairly distributed with reference to gender with females constituting 55.1% and males 44.9% of the total sample size. Students selected in the study belonged to varied income groups to minimize the chances of bias. Most of the respondents believed that on special occasions, more food was getting wasted than on normal days and the consumption stage is the phase where most of the food was being wasted (Table 2).

3.2. Factor analysis

3.2.1. KMO and Bartlett's test

The Kaiser Mayer Olkin's (KMO) test was used to determine whether the data were suitable for factor analysis. The KMO criterion is 0.76, indicating that the sample size is adequate, and the factor analysis is appropriate for the data (Table 3). Factor analysis is typically used to group variables into small clusters based on their common variance. The suitability of the correlation matrix is evaluated using Bartlett's test of sphericity. The correlation matrix contains substantial correlations between the variables, as evidenced by Bartlett's test of sphericity. In our study, Bartlett's test of sphericity value was 7258.15 with a high significance level ($p = 0.001$).

3.2.2. Component matrix

The objective of the PCM is to minimize the dimensionality of the data while retaining as much information as possible by detecting and utilizing the structure in the correlation matrix of the variables included in the study (Taherdoost, 2022). The results suggested that 10 factors may explain 68.6% of the common variation shared by 35 variables (Table 4).

3.2.3. Scree plot

Scree test are used to determine the number of initial unrotated factors to be extracted. In a scree plot, the horizontal axis represents eigenvalue counts, and the vertical axis represents eigenvalue magnitudes. The eigenvalues of the graph are displayed as dots with a line linking each value's subsequent values. Factor extraction should end at the point when the plot levels out or forms an "elbow". In our study, ten

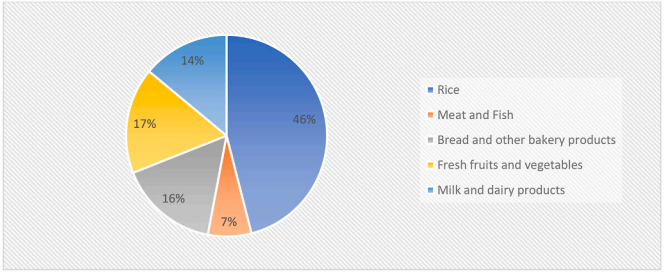


Fig. 2. Food waste percentage depending upon the ingredients.

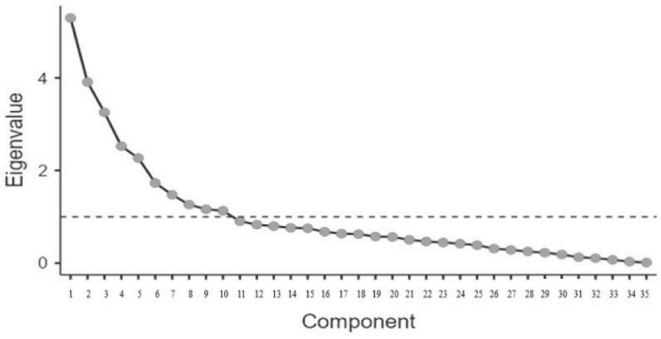


Fig. 3. Scree plot.

Table 2
Descriptive characteristics.

Variables	Description	Frequency (%)
Gender	Male	44.90 %
	Female	55.10%
Age of respondent	18–24	77.70%
	25–30	22.30%
Income of their household	<20,000	16.80%
	21,000–30,000	22.50%
	30,000–50,000	18.50%
	51,000–60,000	17.90%
	Above 60,000	24.30%
Extra foodstuff purchases on special occasions eventually get wasted	less than 10%	34.20%
	10–20 %	18.60%
	50%	47.20%
Students do with leftovers	keep it in the fridge and reuse it	56.00%
	donate it	10.00%
	throw it away	10.00%
	give it to animals and birds	24.00%
The stage at which most of the food is wasted	Production	8.00%
	Transportation	12.00%
	Retail	10.00%
	Consumption	70.00%

Table 3
Kaiser Mayer Olkin's (KMO).

Kaiser-Meyer-Olkin Measure of Sampling Adequacy		.759
Bartlett's Test of Sphericity	Approx. Chi-Square	7258.157
	Df	595
	Sig.	**.000**

Table 4
Principal component matrix.

Component	SS Loadings	% Of Variance	Cumulative %
1	5.22	14.93	14.9
2	2.65	7.57	22.5
3	2.48	7.09	29.6
4	2.43	6.93	36.5
5	2.16	6.17	42.7
6	2.14	6.11	48.8
7	1.87	5.35	54.1
8	1.84	5.25	59.4
9	1.64	4.70	64.1
10	1.58	4.51	68.6

components were selected (Fig. 3).

3.2.4. Rotated component matrix

It is the key output of the principal component analysis. The rotated component matrix was used for a better interpretation of the factors (Table 5). It contains estimates of the correlations between each of the variables and the estimated components (Shrestha, 2021). The results indicate that the purchase behaviour (PB) dominates (Corr. matrix>0.91; $P < 0.05$) in the 1st Component.

3.3. Correlation analysis

The Pearson correlation (r) analysis between various variables indicated less and medium positive correlations (Table 6). Intention not to waste food and moral norms are also moderately positively correlated ($r = 0.419$; $p < 0.01$) and similarly morals norms and knowledge about food waste had moderate correlation ($r = 0.434$). Attitude and purchase behaviour indicated a weak positive correlation among themselves ($r = 0.039$). The correlation between all other variables was also positive (weak and medium) but there was no highly significant correlation between the variables.

3.4. Regression analysis

Regression analysis was performed to determine which factors among the respondents have the most influence on the behavioural intention to not waste food. Hypotheses were tested by using a linear regression model. All hypotheses were supported ($p < 0.01$) except subjective norms that influence their intention to not waste food (Table 7). In other words, subjective norms are not that important to influence the intention ($F = 0.400$, $p > 0.01$). Furthermore, knowledge about food waste has a substantial association with the desire not to waste food ($F = 24.6$, $p < 0.01$, $\beta = 0.651$) implying that students who are aware of food waste and its likely consequences are more likely to minimize food waste.

4. Discussion

Improper disposal of solid waste can lead to serious public health issues and become a nuisance. It can cause the generation of toxic and hazardous compounds that are harmful to both humans and the environment (Adeniyi and Afon, 2022). Nowadays, food waste (FW) is recognized as a critical solid waste that needs to be urgently addressed due to the current conflicts, climate change, hunger crises, and the recent Covid-19 pandemic. Food Wastage is considered as an important CF (Carbon Footprint) that can accelerate and trigger the effects of global warming (Qian et al., 2022a). According to UNEP (2021), FW is the second most important source of carbon emission in the world. Education institutions such as universities are particularly prone to food wastage, which is considered a critical issue (Lazell, 2016; Leal Filho et al., 2021). Thousands of students who reside temporarily in the campus resort to various injudicious feeding behaviours that lead to food wastage (Kaur et al., 2021). In universities, various wasteful activities such as over-purchasing, excessive eating and cooking, improper planning and management, and unorganized food parties are significant predictors of food wastage (Lazell, 2016; Leal Filho et al., 2021). Given the importance of food wastage around the world, especially in universities, our study aimed to explore the core factors responsible for food

Table 5
Rotated component matrix.

Items	1	2	3	4	5	6	7	8	9	10
PB1	.915									
PB2	.925									
PB3	.942									
PB4	.929									
PB5	.942									
PB6	.922									
AT1		.615								
AT2		.637								
AT3		.666								
AT4		.620								
SN1			.850							
SN2			.884							
SN3			.884							
PBC1				.688						
PBC2				.833						
PBC3				.801						
PBC4				.801						
KFW1					.690					
KFW2					.510					
KFW3					.732					
EH1						.633				
EH2						.720				
EH3						.640				
MN1							.707			
MN2							.721			
MN3							.726			
LRR1								.937		
LRR2								.937		
LRR3								.573		
IFW1									.667	
IFW2									.711	
IFW3									.733	
FWB1										.889
FWB2										.890
FWB3										.881

Extraction Method: Principal Component Analysis. Rotation Method: Varimax with Kaiser Normalization.
PB (Purchase Behavior), AT(Attitude), SN (Subjective Norms), PBC (Perceived Behavioral Control), KFW (Knowledge about Food Waste), EH (Eating Habits), MN (Moral Norms), LRR (Leftover Reuse Routine), IFW (Intention not to Waste food), FWB (Positive Food Waste Behavior).

Table 6
Correlation analysis.

	Purchase_Behavior	Attitude	Subjective Norms	Perceived Behavioral Control	Knowledge about Food Waste	Eating Habits	Moral Norms	Leftover Reuse Routine	Intention not to Waste food	Positive Food Waste Behavior
Purchase Behavior	1									
Attitude	.039	1								
Subjective Norms	.020	.055	1							
Perceived Behavioral Control	.058	.038	.250	1						
Knowledge about Food Waste	.030	.339	.034	.048	1					
Eating Habits	.015	.103	.029	.077	.168	1				
Moral Norms	.062	.340	.071	.021	.434	.196	1			
Leftover Reuse Routine	.029	.041	.011	.207	.049	.021	.022	1		
Intention not to Waste food	.040	.295	.034	.100	.340	.149	.410	.026	1	
Positive Food Waste Behavior	.031	.030	.004	.005	.007	.086	.095	.060	.054	1

Table 7
Regression analysis.

Hypotheses	Regression weight	Beta Coefficient	R^2	F	t-value	p-value	Hypotheses Supported
H1	Personal_Attitude → Intention not to waste food	.486	.238	23.58	7.45	.000	Supported
H2	Subjective_Norms → Intention not to waste food	.058	.0.01	.400	7.80	.550	Not-Supported
H3	Perceived_Behavioral_Control → Intention not to waste food	.567	.329	22.25	9.31	.000	Supported
H4	Purchase_Behavior → Intention not to waste food	.436	.228	19.46	2.096	.000	Supported
H5	Knowledge_about_Food_Waste → Intention not to waste food	.651	.364	24.68	10.08	.000	Supported
H6	Moral_Norms → Intention not to waste food	.199	.186	22.398	8.25	.000	Supported
H7	Eating_Habits → Intention not to waste food	.413	.220	7.79	12.12	.006	Supported
H8	Leftover_Reuse_Routine→ Intention not to waste food	.168	.202	16.809	8.316	.000	Supported
H9	Intention not to waste food →Positive_Food_Waste_Behavior	.233	.330	18.135	17.02	.000	Supported

wastage among university students in the Kashmir University, a premier institute in the Northwestern Himalayan region of India. We applied the Extended Theory of Planned Behaviour to uncover these factors. The following discussion is based on the outcomes of the hypotheses.

4.1. Food waste and the role of morals & attitude

The study findings revealed that all the independent variables except subjective norms can significantly predict the intention not to waste food which in turn predicts the positive food waste behaviour among the students. Moral standards, attitudes, and knowledge play a significant role in influencing students food waste behaviour. Moral standards have a big influence and play an essential role in determining students' food waste behaviour. Individuals operate as per their self-concept and waste less food because they feel that wasting food goes against their moral principles and education and negative feelings develop if they throw away food (Aydin and Yildirim, 2020; Stefan et al., 2013; Quested et al., 2013). This study confirms that students who believe wasting food is against their upbringing and moral values follow their self-concept and waste less. Our study aligns with previous research that suggests morality and attitude are key factors that determine whether someone will waste food or not (Stefan et al., 2013; Quested et al., 2013; Richter, 2017). Other studies have shown that an individual's internal moral values regarding food waste play an important role in determining their feeding behaviour (Aydın & Yildirim, 2022). For example, personal norms and behaviours become integrated and internalized within a person's self-concept, and these internalized norms provide essential motivation for individuals to act (Thøgersen, 1996).

It is found that attitude is a significant predictor of intention to avoid food waste, indicating that students often consider ways to reduce food waste (Tsai et al., 2020). Studies have shown that feeling bad about throwing away food is a significant factor in reducing food waste (Richter, 2017). Moral attitudes are affected by the pattern of shopping habits, eating habits, and knowledge of food conservation. Those who believe that wasting food is inherently wrong shop for groceries more prudently, refrain from consuming excess food, and learn ways to preserve food better. Students who have strong moral attitudes against food waste, are more inclined to adopt habits that avoid food waste and are eager to learn about food conservation. Students who were more knowledgeable about food conservation buy more carefully and consider not to buy more than they need. Accordingly, they waste less food (Aydin and Yildirim, 2020). The results of this study are supported by the "self-concept maintenance theory," which suggests that individuals behave in a manner that does not negatively impact their self-image (Mazar et al., 2008). Therefore, individuals who have a strong attitude towards reducing food waste tend to avoid wasting food and learn about food conservation practices (Aydin and Yildirim, 2020).

4.2. Awareness and knowledge about food waste

This research advocates for awareness programs to provide necessary knowledge to the students about food waste problem as knowledge about this significantly impacts the intention not to waste food as confirmed by regression analysis. Meaningful knowing of how to store food properly and avoid waste plays an important role in reducing food waste. Moral norms are correlated with knowledge about food waste and also with the intention to avoid food waste which indicates that if students have more knowledge about food waste problems, their moral

norms align them towards reducing food waste. It is proposed that other types of knowledge, such as effectiveness knowledge (knowledge of the outcomes of one's behavior) or social knowledge (knowledge of other people and social norms) are needed for influencing environmental behaviours (Redman and Redman, 2014). Instead of feeling guilty about throwing away excess food, individuals should take responsibility and realize that by making small changes to their daily eating habits, they can help improve the lives of those who are currently experiencing hunger (Pinto et al., 2018).

It has been found that having knowledge about food wastage is a crucial factor in preventing food wastage, which can lead to significant economic and environmental losses. Learning from past experiences can help improve our understanding of food wastage and minimize it effectively (Farr-Wharton et al., 2014). Prior research has highlighted the significance of having awareness about the environment and food behavior, which are critical aspects for managing food waste (Principato et al., 2015).

Proper knowledge of food preservation is crucial for managing food sustainably in restaurants and eateries located in and around university campuses. By utilizing appropriate preservation methods and techniques, food wastage can be reduced. Research studies have shown that innovative methods and models can have a significant impact on household food waste. It has been established that having accurate knowledge of preservation methods can increase the shelf life of food in households and universities over a specific time period (Janssen et al., 2017). Martindale (2014) also noted that the frozen shelf life of food is associated with reduced food wastage. These studies, along with others, suggest that having proper knowledge of food preparation and use is vital in reducing food wastage (Martindale and Schiebel, 2017).

As pointed out by Knezevic et al. (2019), it is important for university officials and restaurant managers to carry out innovative studies to promote awareness and planning for sustainable food management. The key is to develop holistic mechanisms using multiple initiatives that can disseminate knowledge to all university stakeholders, including students. This will help raise awareness about sustainable food practices and ensure that all members of the university community are informed.

4.3. Perceived and purchasing behaviour

The study's results confirm that perceived behaviour and purchasing habits play a significant role in determining food waste among university students. The theory of Planned Behaviour suggests that people feel more in control when their behaviour is more predictable (Aydin and Aydin, 2022). Feeding habits are a crucial factor in determining food waste behaviour (Russell et al., 2017), which is directly or indirectly related to perceived behaviour. For example, shopping behaviour has been found to be an important factor in determining the amount of food wasted by households (Stefan et al., 2013). It has been suggested that perceived behavioural control can be used to predict behavioural intention and actual behaviour (Ajzen, 2011). These behaviours are essential for one's intention to carry out a particular behaviour, such as food waste behaviour.

Purchasing or shopping behaviour is considered an important factor that is responsible for food wastage. Well-managed and multiple purchasing behaviours can facilitate food wastage in households, public places and university institutions (Ahmed et al., 2021). Shopping behaviour, such as planning ahead for shopping based on needs and estimates, could help less shopping or avoid trash shopping, which could lead to less shopping and ultimately lead to the conservation of food (Ahmed et al., 2021).

4.4. Eating habits

Food wastage is a crucial issue in universities where students from diverse backgrounds adopt different eating habits. Research studies have shown that positive cultural traditions related to eating habits can help reduce food wastage, as observed in Turkey (Schanes et al., 2018; Surucuoglu, 2008). The habit of finishing the food that is served on the plate depends on the innate tendency of the students or consumers. Additionally, the abundance of food available on the plate can also lead to food wastage (Porpino et al., 2016). Therefore, eating habits alone are not solely responsible for food wastage. Cultural activities and the availability of food also influence feeding habits. For instance, in some Latin American cultures, people prepare more food than they eat, resulting in a vast amount of food being served on the table and wasted (Porpino et al., 2016).

4.5. Food waste and social norms

In our study, we found that food is wasted at every step of the food supply chain but the consumer stage typically makes up the largest portion of the overall food waste (Van Alboom, 2021). If we need to solve the issue of food waste, it is this stage which needs utmost attention, and it can be certainly achieved by particularly focusing on the behavioral aspects of the target group. Out of the five product categories, rice emerged as the most wasted food item and meat/fish the least. The possible reason is that rice is the staple food in Kashmir and is readily available. This is also supported by Favis et al. (2022) who concluded that rice is the overall dominant food waste category in the educational institutes of the Asian countries. Meat and fish are expensive and are not eaten regularly, most students cannot afford them and eventually are found less in quantity as food waste. Furthermore, food is wasted on special occasions like marriages, birthday parties, and get-togethers as extra food is prepared on these occasions. Some students believed that they were using leftover foods to avoid food waste, but these formed only a small portion of the students. It is advocated if students are encouraged to store food in the refrigerators, we can reduce the food waste considerably.

The results of this study have consequences for educational institutions and families. Socialisation agents like parents and teachers have a significant impact in defining an individual's moral standards during the formative stages of life (Narvaez and Lapsley, 2009). Therefore, providing students with moral role models who can help them develop good moral attitudes can be highly beneficial in preventing food waste.

5. Conclusions

In this research, we have offered a framework for understanding food waste behaviour among the young generation which will help in comprehending food waste management and prevention of food waste. There are a variety of elements, such as one's attitude, behavioural control, knowledge about food waste, moral standards, eating habits, and leftover reuse routine which significantly affect one's desire not to waste food. Subjective norms did not reach statistical significance and could therefore not be used to predict intentions. As most of the food waste is generated at the consumption stage which predominantly depends on students' behavior and the choices they make, we can target the factors that directly influence one's food waste behavior. The study's findings give a deeper understanding of these influences on FW by demonstrating the mediating roles they play in describing food waste behaviour. Furthermore, the results confirm prior research that socio-demographic characteristics are only weak predictors of behaviour related to food waste. The current study offers an alternative perspective on food waste incorporating a behavioural approach rather than focusing on socio-demographics which have little impact on remedies and solutions. The study unequivocally identifies highly effective strategies that can significantly reduce food waste among young people. These strategies outline the most impactful factors that explain the intention to waste less food, which can make a tangible contribution to addressing the global issue of food waste.

6. Directions for future research

Reducing food waste is an absolute social responsibility of universities, and they must prioritize enhancing their environmental stewardship. This will not only improve their public image but also attract students who are passionate about sustainability programs. To transform students' education, universities must involve them in the sustainability education process through practical and cooperative approaches to address food waste. However, this is only achievable if more research is carried out to identify effective ways to reduce or eliminate food waste. It is imperative that universities function as living labs to bring new approaches and development ideas for feasible solutions that can assist with the challenges posed by food waste (FW). Sustainable methods of FW management must be identified and implemented urgently, and universities, as microcosms of learning and innovation, can play an active and crucial role in reducing and/or recycling FW. However, this effort requires systemic and strategic actions because it is a complex task that implies monitoring and controlling waste against set targets.

An on-campus composting facility at a university and community-based composting can be a catalyst for generating innovative solutions to reduce emissions and promote sustainability. The knowledge gained from this effort can be shared and implemented on a larger scale, leading to a positive impact on the environment. Therefore, it is crucial to ensure that the composting process meets the required standards to produce an effective and high-quality product. By doing so, we can contribute to a more sustainable future for our planet.

Even though many individuals waste food, studies on what discourages people from doing so need to be more consistent. Although addressing younger individuals and students in various locations may be advantageous for food waste reduction initiatives, methods should be created to target food-related behaviours among the masses that are responsible for food waste.

CRediT authorship contribution statement

Shahnaza Akhter: Conceptualization, Data curation, Formal analysis, Investigation, Validation, Visualization, Writing – original draft. **Mohmmad Irshad Rather:** Conceptualization, Data curation, Formal analysis, Methodology, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Ummer Rashid Zargar:** Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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